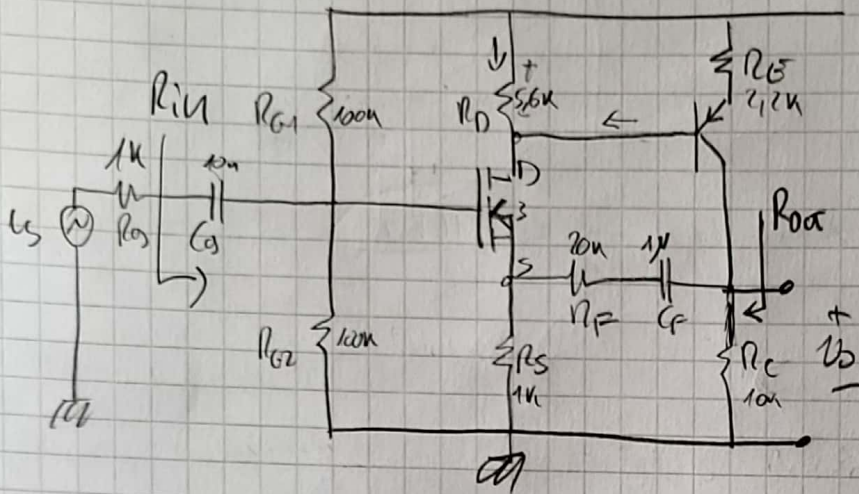


BS 1

$V_{DD} = 9V$



$$I_{RD} = I_D - I_B$$

1) DC point

$$V_{GS} = 4.5V \quad R_{GS} = 50k\Omega$$

$$\begin{cases} V_{GS} - V_{GS} - R_S I_D = 0 \\ I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_T}\right)^2 \end{cases} \Rightarrow \begin{cases} I_D = \frac{V_{GS} - V_{GS}}{R_S} \\ \frac{V_{GS} - V_{GS}}{R_S} = I_{DSS} \left(1 - \frac{V_{GS}}{V_T}\right)^2 \end{cases}$$

$$\frac{V_{GS}}{R_S I_{DSS}} - \frac{V_{GS}}{R_S I_{DSS}} = 1 + \frac{V_{GS}^2}{V_T^2} - \frac{2V_{GS}}{V_T}$$

$$V_{GS}^2 \left(\frac{1}{V_T^2}\right) + V_{GS} \left(-\frac{2}{V_T} + \frac{1}{R_S I_{DSS}}\right) + 1 - \frac{V_{GS}}{R_S I_{DSS}} = 0 \quad \boxed{V_{GS} = 4.16V}$$

$$\boxed{I_D = 0.336mA}$$

$$-R_D(1+\beta_F)I_B + V_{BE} + R_D I_{RD} = 0$$

$$-R_D(1+\beta_F)I_B + V_{BE} + R_D I_D - R_D I_B = 0$$

$$I_B = \frac{V_{BE} + R_D I_D}{(1+\beta_F)R_D + R_D} = 5.18 \mu A \quad \boxed{I_C = 0.518mA}$$

$$V_{DS} = V_{CC} - R_D(I_D - I_B) - R_S I_D = 6.81V$$

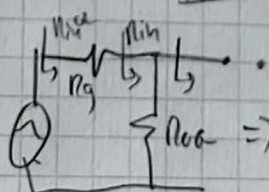
$$V_{CE} = V_C - V_E = R_C I_C - (V_{CC} - R_D I_E) = -2.67V$$

$$g_{m1} = 0.580 \text{ mS}^{-1}$$

$$g_{m2} = 20.72 \text{ mS}^{-1}$$

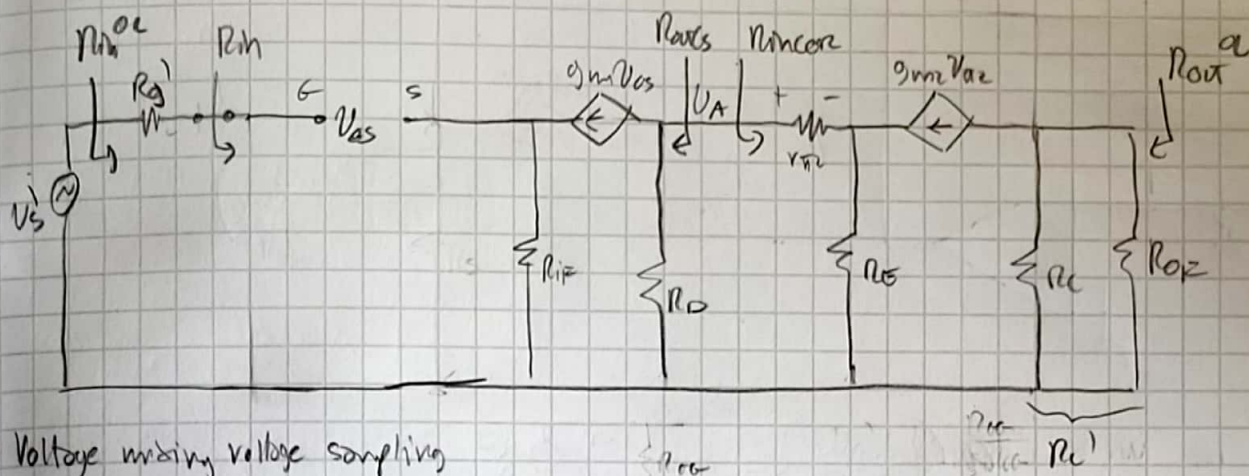
$$V_{T1} = 4.82 \text{ k}\Omega$$

Therem on input

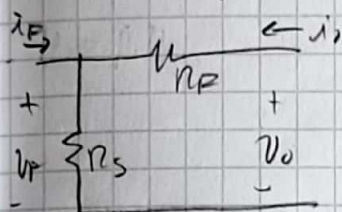


$$\frac{V_o}{V_{gs}} = \frac{R_{GS}}{R_S + R_{GS}} \quad R_{GS} = R_{GS} / R_{GS}$$

$$2) A_V = V_o / V_s$$



Voltage mixing voltage sampling



$$\begin{cases} V_F = \beta V_o + R_{EF} i_F \\ i_o = V_o / R_{EF} \end{cases}$$

$$\begin{cases} \beta = \frac{V_o}{V_F} = \frac{R_E}{R_E + R_B} = 0,0976 \\ R_{EF} = R_E // R_E = 0,952 \text{ k}\Omega \\ R_{OF} = R_E + R_E = 2 \text{ k}\Omega \end{cases}$$

$$A_V = \frac{V_o}{V_g} = \frac{V_o}{V_A} \cdot \frac{V_A}{V_g}$$

$$\frac{V_o}{V_A} = \frac{-\beta R_C}{R_{in} + R_C} = -2,94$$

$$\frac{V_A}{V_g} = -g_m (R_C // R_{in}) \cdot \frac{1}{1 + g_m R_{EF}} = -2,09$$

$$R_C' = 6,67 \text{ k}\Omega$$

$$R_{in} = R_B // R_E (1 + \beta) = 227 \text{ k}\Omega$$

$$R_{out} = R_C$$

$$A_V^{OL} = 5,94 \quad A_V^F = \frac{A_V^{OL}}{1 + A_V^{OL} \beta} = 4,66$$

$$A_V = \frac{V_o}{V_g} = A_V^F \cdot \frac{V_g}{V_A} = 4,57$$

3) R_{in}

$$R_{in}^{OL} = R_{in}^F = \infty \quad (R_{in}^F - R_g) // R_{OG} = R_{OG} = 50 \text{ k}\Omega$$

4) R_{out}

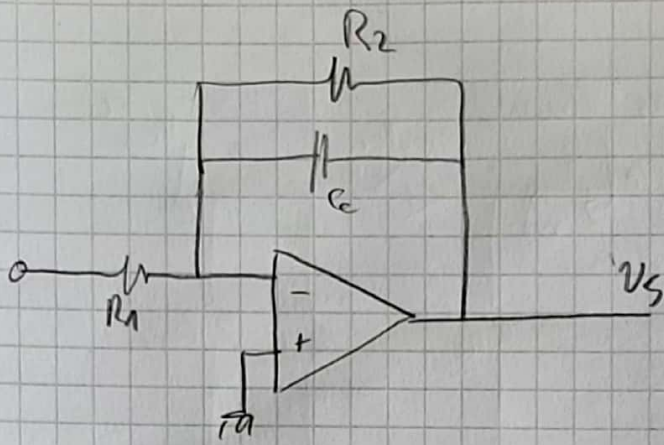
$$R_{out}^{OL} = 6,67 \text{ k}\Omega = R_C' \quad R_{out}^F = R_{out}^{OL} // 11 \text{ k}\Omega = 5,18 \text{ k}\Omega$$

5) f_L considering only C_g

$$R_{g^x} = 51 \text{ k}\Omega \quad f_L^a = f_L^F = \frac{1}{2\pi} \left(\frac{1}{C_g R_{g^x}} \right) = 312 \text{ Hz}$$

for you can apply miller to C_F

652



$$A(s) = \frac{A_{20}}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$

1) estimate β_m without C

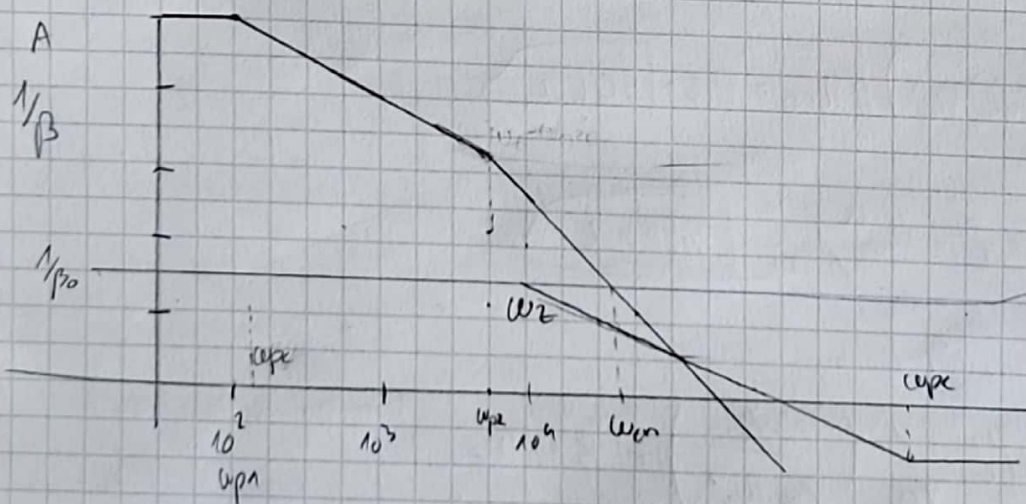
$$T = \frac{v_r}{v_e} = A(s) \cdot \frac{z_1}{z_2 + z_1} \quad z_1 = R_1 \quad z_2 = R_2 \parallel \frac{1}{sC} = \frac{R_2}{1 + sC R_2}$$

$$= A(s) \cdot \frac{R_1}{\frac{R_1 + \frac{R_2}{1 + sC R_2}}{1 + sC R_2}} = \frac{R_1}{\frac{R_1(1 + sC R_2) + R_2}{1 + sC R_2}} = \frac{R_1(1 + sC R_2)}{(R_1 + R_2)(1 + sC(R_1 \parallel R_2))} \quad A(s)$$

$$\beta_0 = \frac{R_1}{R_1 + R_2} = \frac{1}{36} \quad \frac{1}{\beta_0} = 36 = 30,6 \text{ dB}$$

$$\beta_{\infty} = \frac{R_2}{R_1 \parallel R_2} = 36 \quad \frac{1}{\beta_{\infty}} = \frac{1}{36} = -30,6 \text{ dB}$$

$$\omega_{p1} = \frac{1}{C(R_1 \parallel R_2)} \gg \omega_2 = \frac{1}{C R_2}$$

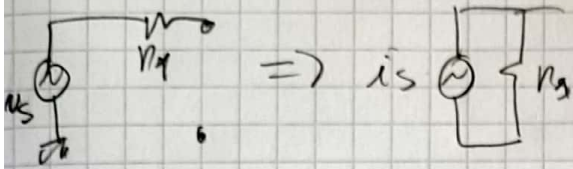


$$\alpha = 40 \text{ dB} \rightarrow \phi \rightarrow 0^\circ$$

$$\begin{cases} A_{20} \omega_{p1} = A_{\infty} \omega_{p2} \\ A_{\infty} \omega_{p2}^2 = \frac{1}{\beta_0} \omega_{en}^2 \end{cases} \quad \sqrt{A_{20} \omega_{p1} \omega_{p2} \beta_0} = \omega_{en} = 3,83 \cdot 10^4 \text{ rad/s}$$

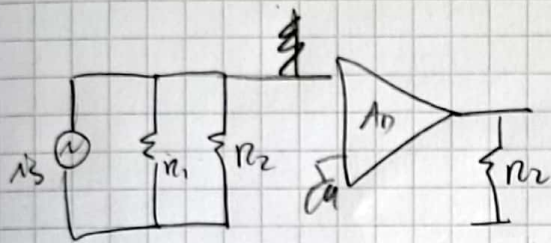
$$\varphi_m = 180 - \arctan\left(\frac{\omega_{ce}}{\omega_{p1}}\right) - \arctan\left(\frac{\omega_{ce}}{\omega_{p2}}\right) = 7,57^\circ$$

2) BD deriving to feedback theory (with C_c open)
current mixing voltage sampling

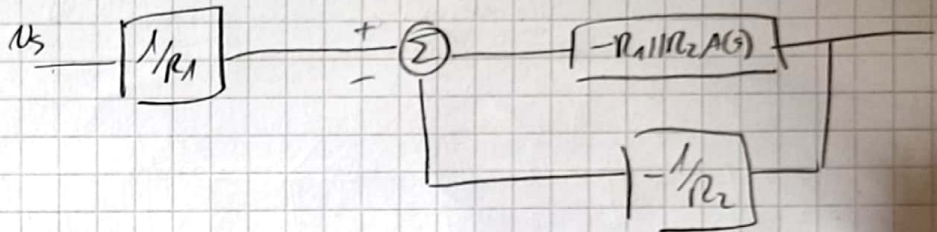


$$K_s = 1/R_1$$

$$\beta = -1/R_2 \quad R_{ip} = R_{os} = R_2$$



$$A_{be} = \frac{v_o}{i_s} = -A_o(s) R_1 // R_2$$



3) C_c to stabilize the amplifier

We have to move ± 20 dB the opening/closing ratios

introducing C_c we introduce 0 zero and 2 pole we put the zero 2 octaves before ω_{ce}

$$\omega_z = \frac{1}{C_c R_2} = \frac{\omega_{ce}}{G}$$

$$C_c = \frac{G}{\omega_{ce} R_2} = 3,16 \mu F \rightarrow \varphi_m + 60^\circ$$

$$C_c = \frac{1}{\omega_{ce} R_2} = 790 \text{ pF} \rightarrow \varphi_m + 45^\circ$$

4) draw frequency

~~Sketch~~

16:05 Starto Simul